MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6E5

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electron-ray tube, indicator type

Outline	---	9-26 (EIA) Ⓐ
Base	---	A6-7
Envelope	---	T9
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6
Element	---	h	p (see note 1)	g	ta	k (see note 1)	h

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Ebb	Ec	Eta	Ehk	Alt
Unit:	V	Vdc	Vdc	Vdc	v	ft
Maximum:	6.9	275	---	275	100	10,000
Minimum:	5.7	---	---	125	---	---
TEST CONDITIONS:	6.3	250	-4	250	---	---

REQUIREMENT OR TEST:

GENERAL

3.2	Qualification - Required	E-50.3	Preheating Ⓐ
E-50.2	Holding period	5.	Preparation for delivery

Ⓐ denotes changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:51Z
Check the source to verify that this is the current version before use.

METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1							
1266	Total grid current	See note 6	0.65	II	Ic	0	-1.5	μAdc
1256	Target current		0.65	II	Ita	0.2	6.0	mAdc
---	Pattern check (1)	Ec = -6 Vdc; Rp = 1 Meg (see note 2)	0.65	II	---	---	---	---
---	Pattern check (2)	Ec = -10.5 Vdc; Rp = 1 Meg (see note 3)	0.65	II	---	---	---	---
1231	Emission	Eb = Eta = Ec = 30 Vdc (see note 6)	0.65	II	Is	15	---	mAdc
1201	Short and discontinuity detection		0.4	II	---	---	---	---
Appendix D. 20(a), 30	Visual and mechanical inspection criteria		---	---	---	---	---	---
Appendix D. 20(b)	Dimensions	Outline 9-26; dimensions A thru D	---	---	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		4.0	L6	R	10	---	Meg
1031	Low-frequency vibration	Ebb = 100 Vdc; Ec = -1 Vdc; Rp = 10,000 ohms; Eta = 100 Vdc	6.5	L6	Ep	---	150	mVac
1301	Heater current		6.5	L6	If	275	325	ma
Ⓐ 1336	Heater-cathode leakage		6.5	L6	Ihk	---	20	μAdc
1256	Anode current		6.5	L6	Ib	3.4	7.2	mAdc
---	Light output	Ec = -10.5 Vdc; Rp = 1 Meg (see note 5)	6.5	L6	Light	4.0	---	ftC
1101	Secureness of base and base insert		6.5	L6	---	---	---	---
1111	Base pin solder depth		6.5	L6	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
4.7	Life-test provisions (100 hours)	Group A; Rp = 1.0 Meg; Ehk = 100 V; Ec/pattern close	---	---	---	---	---	---
4.7.3	Life-test end points (100 hours)	Target current (see note 4)	---	---	Ita	---	6.5	mAdc
		Light output (see note 5)	---	---	Light	2.5	---	ftC

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:51Z
Check the source to verify that this is the current version before use.

NOTES:

1. The shadow electrode shall fall within ±12° of a line parallel to a plane passing through the base pins for the anode and cathode (pin numbers 2 and 5), and shall be located over the cathode pin (pin number 5).

2. The pattern shall not meet for more than half of radial width of pattern.

3. The pattern shall meet at some point. Tubes having images which show dark areas, sponginess, or poor illumination shall be rejected.

4. If the tube fails to meet either of the pattern check tests, it will be considered a defective tube.

5. The light output shall exceed the limit specified when measured with a foot-candle meter, corrected for the spectral response of the eye, placed in direct contact with the top of the tube. All remaining area of meter and tube shall be blanked off to prevent leakage of external light. The meter reading shall be corrected for the area of the meter exposed to the light from the tube.

6. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

(Project 5960-2199-3)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:51Z
Check the source to verify that this is the current version before use.